
Multi Threading

Multi Threading is a process of running multiple tasks simultaneously at the same time.

To work on this, we need to place each task in different threads.

Syntax:

```
import threading  
t1 = threading.Thread(target = TaskName)
```

Functions Used in Threads

1. start()

- It is used to start the execution of a thread

2. join()

- It is used to join the execution of multiple threads

3. is_alive()

- It returns True if the thread is active
- It returns False if the thread is not active

4. getName()

- It is used to get the name of a thread

5. setName()

- It is used to set/change the name of a thread
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Example:

```
import time
import threading

start = time.time()

def something():
    for i in range(5):
        print("Do something")
        time.sleep(1)

def nothing():
    for i in range(5):
        print("Do nothing")
        time.sleep(1)

t1 = threading.Thread(target=something)
t2 = threading.Thread(target=nothing)

# Default thread names
print("Default Name t1:", t1.getName())
print("Default Name t2:", t2.getName())

# Setting new names
t1.setName("King")
t2.setName("Queen")

# New thread names
print("New Name t1:", t1.getName())
print("New Name t2:", t2.getName())
```

is_alive before start

```
print("Is t1 alive before start?", t1.is_alive())  
print("Is t2 alive before start?", t2.is_alive())
```

```
t1.start()
```

```
t2.start()
```

is_alive after start

```
print("Is t1 alive after start?", t1.is_alive())  
print("Is t2 alive after start?", t2.is_alive())
```

```
t1.join()
```

```
t2.join()
```

```
end = time.time()
```

```
print("Total time taken =", end – start)
```
